Foundations of Generative AI (ICT4403)
Assignment 3 Submission Document

Neural Machine Translation System (English to French) using Basic Transformer, mT5, LLaMA, and Gemma

1. Submission Details

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Assignment Title	Foundations of Generative AI (ICT4403) - Assignment 3
Title of Developed System	Comparative English-to-French Neural Machine Translation System using Encoder-Decoder (Basic Transformer, mT5) and Decoder-Only (LLaMA, Gemma) Architectures
Dataset Name and Link	Tatoeba / OPUS English-French Parallel Corpus https://www.manythings.org/anki/fra-eng.zip | https://huggingface.co/datasets/Helsinki-NLP/opus-100
GitHub Repository Link	https://github.com/Kshiku12/Gen-AI-Assignment

2. Executive Summary & Quantitative Evaluation

This project develops, adapts, and empirically evaluates an English-to-French (en -> fr) Neural Machine Translation system across four distinct generative architectures: (1) a Basic Encoder-Decoder Transformer trained from scratch in PyTorch, (2) mT5-small (google/mt5-small, Encoder-Decoder) fine-tuned via supervised Seq2Seq learning, (3) LLaMA-3.2-1B-Instruct (Decoder-Only) adapted via both 3-Shot In-Context Prompting and LoRA Fine-Tuning, and (4) Gemma-3-1B-IT (Decoder-Only) adapted via both 3-Shot In-Context Prompting and LoRA Fine-Tuning. All models were evaluated on the identical held-out test set from the Tatoeba English-French Parallel Corpus (12,000 sentence pairs: 9,600 train, 1,200 validation, 1,200 test).

Model	Architecture	Adaptation	BLEU	chrF	Train (s)	Latency
Basic Transformer	Encoder-Decoder	Trained from Scratch	22.53	42.55	68.0s	1.2 ms
mT5-small	Encoder-Decoder	Full Seq2Seq Fine-Tuning	31.28	51.07	532.7s	35.3 ms
LLaMA-3.2-1B (Prompting)	Decoder-Only	3-Shot In-Context Prompting	34.68	56.18	0.0s	166.6 ms
LLaMA-3.2-1B	Decoder-Only	LoRA Fine-Tuning (r=16, alpha=32)	38.43	57.09	168.4s	96.4 ms
Gemma-3-1B (Prompting)	Decoder-Only	3-Shot In-Context Prompting	44.88	62.60	0.0s	400.1 ms
Gemma-3-1B	Decoder-Only	LoRA Fine-Tuning (r=16, alpha=32)	47.47	63.08	215.8s	177.9 ms

3. Key Architectural, Temperature & Ethical AI Findings

• **Encoder-Decoder vs. Decoder-Only Comparison:** The from-scratch Basic Transformer achieves ultra-low inference latency with a compact parameter footprint (6.27M parameters), whereas pretrained multilingual models (mT5-small,

LLaMA-3.2-1B, and Gemma-3-1B) leverage massive subword vocabularies and web-scale pretraining to handle rare words, complex syntax, and idioms with substantially higher BLEU and chrF scores.

• **Impact of Adaptation Method (Prompting vs. LoRA):** Applying parameter-efficient LoRA fine-tuning (updating <0.3% of parameters) to LLaMA-3.2-1B and Gemma-3-1B aligns their outputs to target domain conciseness and French punctuation conventions, significantly boosting BLEU and chrF over 3-shot prompting while eliminating multi-shot prompt overhead during inference.

• **Sampling-Temperature Analysis:** Greedy decoding (T = 0.0) and low temperature (T = 0.3) maximize translation fidelity (BLEU and chrF), whereas higher temperatures (T = 0.7, 1.0) increase lexical diversity (Distinct-2) at the cost of semantic drift.

• **Responsible & Ethical AI Audit:** Diagnostic testing reveals occupational gender bias when translating gender-neutral English roles into French (defaulting 'doctor' and 'engineer' to masculine 'le médecin'/'l'ingénieur' and 'nurse'/'secretary' to feminine 'l'infirmière'/'la secrétaire'). Mitigation measures including counterfactual gender data augmentation, terminology-constrained decoding, and dual-gender templates are proposed.

• **Best-Performing Model:** Gemma-3-1B is identified as the best-performing translation system (BLEU: 47.47, chrF: 63.08).